

3F7: Information Theory and Coding

Handout 3: Prefix-free codes, Kraft inequality, Lossless source coding theorem

Ramji Venkataramanan

rv285@cam.ac.uk

Michaelmas Term 2025

1 / 21

Morse Code

A	●—	J	●— — —	S	● ● ●
B	— ● ● ●	K	— ● —	T	—
C	— ● — ●	L	● — ● ●	U	● ● —
D	— ● ●	M	— —	V	● ● ● —
E	●	N	— ●	W	● — —
F	● ● — ●	O	— — —	X	— ● ● —
G	— — ●	P	● — — ●	Y	— ● — —
H	● ● ● ●	Q	— — ● —	Z	— — ● ●
I	● ●	R	● — ●		

The Morse code assigns a binary codeword to each letter: think of ● as 0 and — as 1.

- ▶ Note that more frequent letters such as *A*, *E* have shorter codewords and rarer letters like *J*, *Z* have longer codewords.
- ▶ But, given a code sequence ● — — ● ●, what is the corresponding English sequence? Could be *ATI* or *EGE* or *EZ*
- ▶ So in addition to ● and —, Morse code has two additional symbols for *letter space* and *word space*.

Can we design a better code that allows decoding without these spaces?

2 / 21

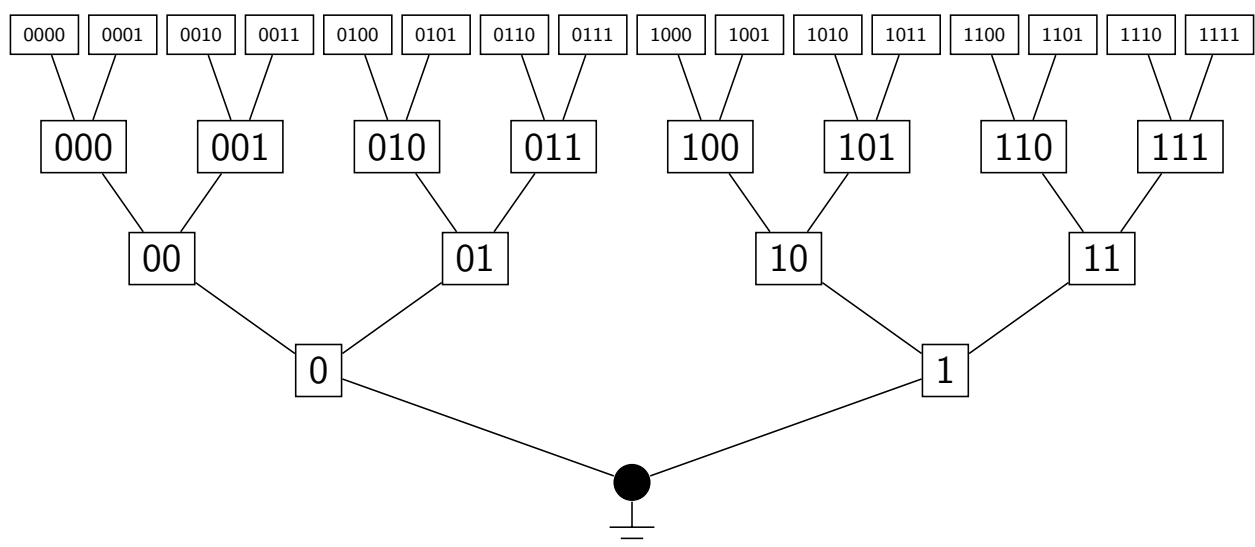
The codeword supermarket

0	00	000	0000	...
			0001	...
		001	0010	...
			0011	...
	01	010	0100	...
			0101	...
		011	0110	...
			0111	...
1	10	100	1000	...
			1001	...
		101	1010	...
			1011	...
	11	110	1100	...
			1101	...
		111	1110	...
			1111	...

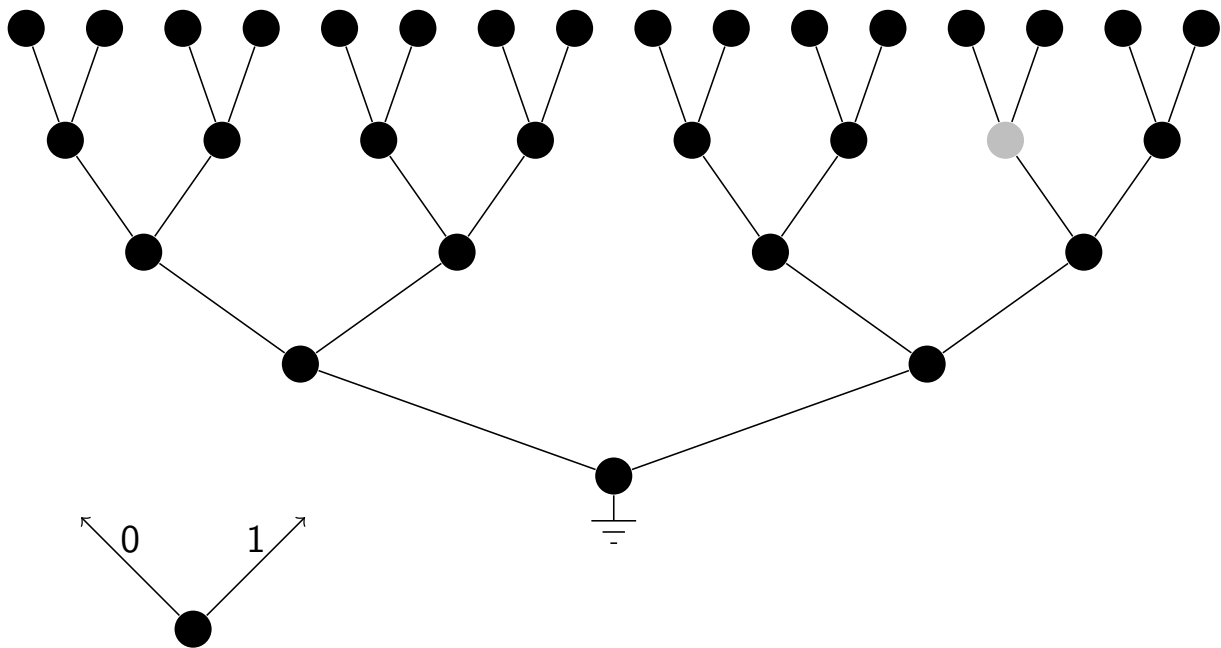
This table lists all available codewords, i.e., all possible binary sequences of length 1, then length 2, ...

3 / 21

The codeword supermarket on a tree



4 / 21



We can remove the labels if we follow the convention that a move to the left adds a 0, and a move to the right adds a 1.

What is the binary sequence corresponding to the grey node?

5 / 21

Prefix-free codes

A code is called *prefix-free* or *instantaneously decodable* if no codeword is the prefix of another.

Which of the following codes is prefix-free?

1. $\{0, 00, 10\}$?
2. $\{0, 10, 11\}$?
3. $\{00, 010, 011, 0101, 1\}$?

Assuming the source alphabet is $\{a, b, c\}$, what is the decoded string for 00010 with: a) code 1, and b) code 2?

In a prefix-free code, each symbol can be decoded as soon as you come to the end of the codeword corresponding to it. There is no need for letter spaces or word spaces.

6 / 21

Extension Codes and Unique Decodability

Given a binary code C for alphabet $\mathcal{X}$, the extension code C^n is the code applied symbol-by-symbol to strings in $x^n \in \mathcal{X}^n$, i.e., the codeword for x^n is

$$C(x_1x_2 \dots x_n) = C(x_1)C(x_2) \dots C(x_n)$$

The extension code C^n is called *uniquely decodable* if for each binary codeword in C^n , there is only one possible source string that can produce it.

Prefix-free codes are uniquely decodable, but not all uniquely decodable codes are prefix-free:

When n is large you may be able to uniquely decode the source string by looking at the entire binary codeword in C^n , even though you may not be able to decode instantaneously symbol-by-symbol.

7 / 21

We will focus only on designing and analysing prefix-free codes as we want fast encoding and decoding algorithms.

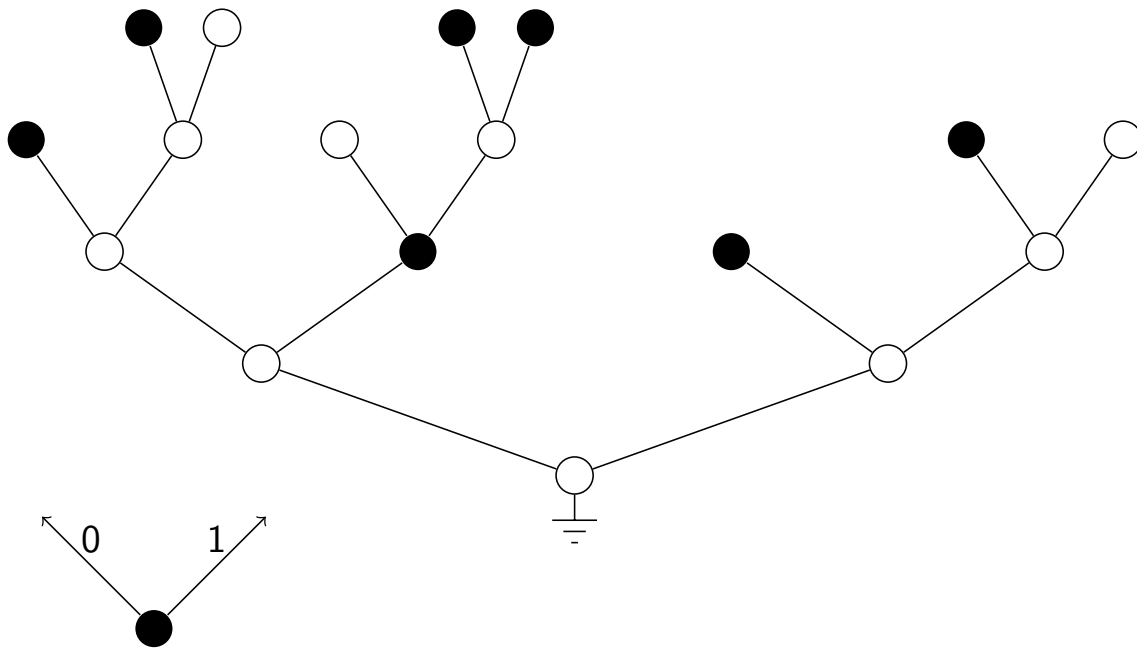
Recall that for a source with alphabet $\mathcal{X}$ and the probability of each symbol given by pmf P , a code that assigns a codeword of length $\ell(x)$ to each symbol $x \in \mathcal{X}$, has expected length

$$\sum_{x \in \mathcal{X}} P(x)\ell(x).$$

We wish to construct prefix-free codes of minimum expected length to describe a given source.

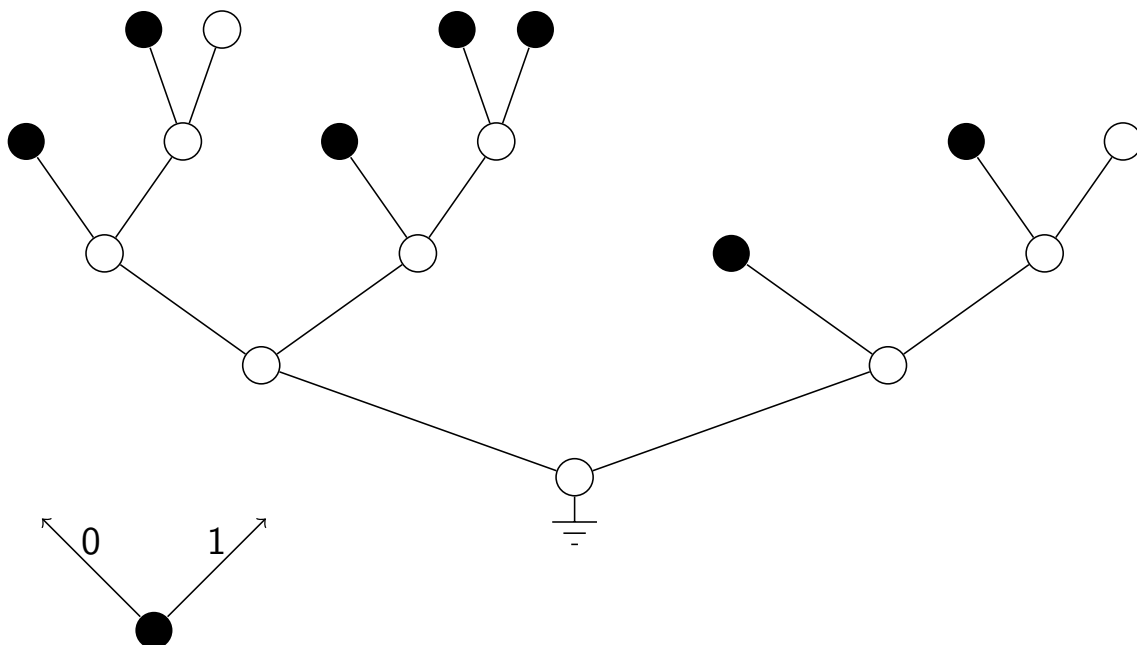
8 / 21

Prefix condition on a tree



- This code with codewords $\{000, 0010, 01, 0110, 0111, 10, 110\}$ is *not* prefix-free

9 / 21



- This code $\{000, 0010, 010, 0110, 0111, 10, 110\}$ is prefix-free.
- Prefix-free codes have codewords on the **leaves** of the tree.
- The codewords of this code have lengths $\{\ell_1, \ell_2, \ell_3, \ell_4, \ell_5, \ell_6, \ell_7\} = \{3, 4, 3, 4, 4, 2, 3\}$

10 / 21

Prefix code lengths

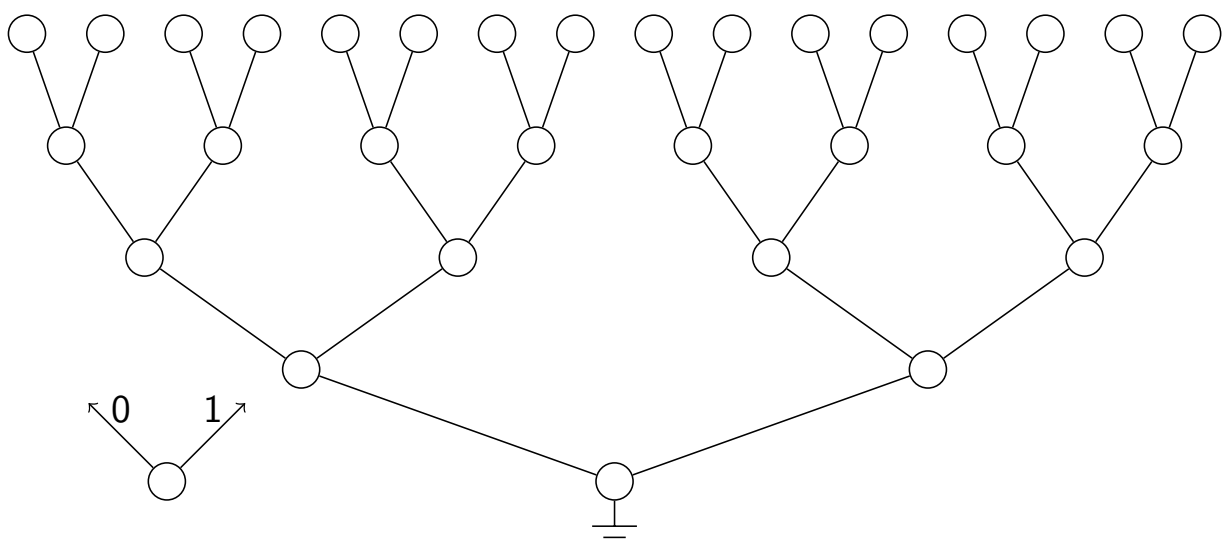
Do binary prefix-free codes with the following word lengths exist?

- ▶ $\{1, 2, 3, 4, 4\}$
- ▶ $\{2, 2, 2\}$
- ▶ $\{2, 2, 2, 2, 2\}$
- ▶ $\{1, 2, 4, 4, 5, 5, 5, 6, 7, 7, 7\}$

This is equivalent to asking if we can construct trees with leaves at these specified depths.

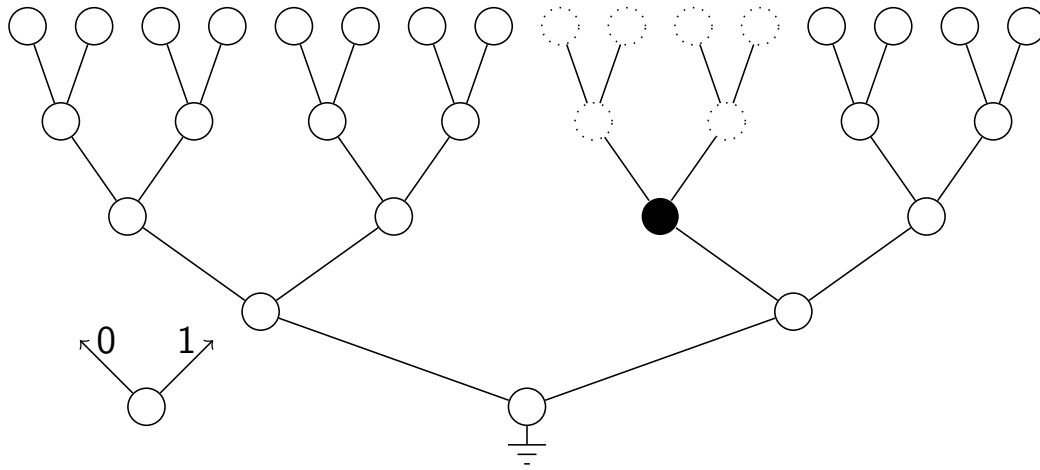
11 / 21

Pruning the tree



- ▶ Full tree of depth $\ell_{\max}$ has $2^{\ell_{\max}}$ leaves

12 / 21



- ▶ Whenever a codeword of length $\ell < \ell_{\max}$ is created, all its descendants are removed from the tree
- ▶ In particular, each codeword of length ℓ leads to a set of $2^{\ell_{\max}-\ell}$ **unusable leaves** at depth $\ell_{\max}$. Also, the sets of unusable leaves created by the codewords are non-overlapping.
- ▶ Hence the total number of unusable leaves at level $\ell_{\max}$ for a code with word lengths $\{\ell_1, \ell_2, \dots, \ell_N\}$ is $\sum_{i=1}^N 2^{\ell_{\max}-\ell_i}$

13 / 21

Kraft Inequality

Since the total number of leaves at level $\ell_{\max}$ is $2^{\ell_{\max}}$, the argument on the previous page shows that

$$\sum_{i=1}^N 2^{\ell_{\max}-\ell_i} \leq 2^{\ell_{\max}} \Rightarrow \sum_{i=1}^N 2^{-\ell_i} \leq 1.$$

This is the Kraft inequality, which we have shown is a *necessary* condition for any prefix-free code with lengths $\{\ell_1 \dots \ell_N\}$.

The Kraft inequality is also a *sufficient* condition, i.e., if a set of lengths satisfy it, we can always construct a prefix-free code with these lengths. Indeed, label the first node of depth ℓ_1 (from the left) as codeword 1, and eliminate its descendants. Then label the first remaining node of depth ℓ_2 as codeword 2, and so on.

Summarizing, we have shown the following important result.

A binary prefix-free code with codeword lengths $\ell_1, \ell_2, \dots, \ell_N$ exists if and only if

$$\sum_{i=1}^N 2^{-\ell_i} \leq 1$$

14 / 21

In fact, the Kraft inequality is satisfied by any *uniquely decodable* code, not just prefix-free codes:

If C^n is any uniquely decodable (extension) code which assigns a binary codeword of length $\ell(x^n)$ to each $x^n \in \mathcal{X}^n$, then

$$\sum_{x^n} 2^{-\ell(x^n)} \leq 1.$$

We will not prove this result here, but if interested you can find the proof in [Cover & Thomas, Sec. 5.5].

15 / 21

Assigning codewords to source symbols

- ▶ We now know which set of codeword lengths are feasible, and which are not.
- ▶ Given a feasible set of codeword lengths, we want to minimize the expected codeword length L given by

$$L := \mathbb{E}[\ell(X)] = \sum_{x \in \mathcal{X}} p(x) \ell(x)$$

- ▶ We also want to answer questions like: how close to optimal (in terms of expected code length) is the following code:

Source symbol	Probability	Codeword	Codeword length
A	0.15	000	3
B	0.07	0010	4
C	0.17	010	3
D	0.06	0110	4
E	0.06	0111	4
F	0.31	10	2
G	0.18	110	3

16 / 21

Coding theorem for a random variable

Let X be a random variable taking values in $\mathcal{X}$ with entropy $H(X)$. The expected codeword length L of any binary prefix-free code for X satisfies

$$L \geq H(X).$$

Proof. We show that $H(X) - L \leq 0$ as follows. Denote the probabilities of the symbols as $p_1, p_2, \dots$, and the corresponding codeword lengths as $\ell_1, \ell_2, \dots$.

Then:

$$\begin{aligned} H(X) - L &= \sum_i p_i \log_2(1/p_i) - \sum_i p_i \ell_i \\ &= \sum_i p_i \log_2 \frac{2^{-\ell_i}}{p_i} = \frac{1}{\ln 2} \sum_i p_i \ln \frac{2^{-\ell_i}}{p_i} \\ &\stackrel{(a)}{\leq} \frac{1}{\ln 2} \sum_i p_i \left(\frac{2^{-\ell_i}}{p_i} - 1 \right) \\ &= \frac{1}{\ln 2} \left(\sum_i 2^{-\ell_i} - \sum_i p_i \right) \\ &\stackrel{(b)}{\leq} 1 - 1 = 0. \end{aligned}$$

In the above chain of inequalities, (a) is obtained using the inequality $\ln(x) \leq x - 1$, for all $x > 0$.

(b) is due to Kraft's inequality.

Coding theorem for blocks of source symbols

Instead of assigning a codeword to each source symbol, suppose that you have any prefix-free code that assigns binary codewords to *blocks* of N source symbols.

Denoting the block by $X^N := (X_1, \dots, X_N)$, the above theorem can be applied to the code for X^N . (Think of the block X^N as a random variable taking values in the larger space $\mathcal{X}^N$.)

Hence the expected length of the code satisfies

$$\mathbb{E}[\ell(X^N)] \geq H(X^N) \quad (\text{or}) \quad \frac{\mathbb{E}[\ell(X^N)]}{N} \geq \frac{H(X^N)}{N}$$

$\frac{\mathbb{E}[\ell(X^N)]}{N}$ is the average number of code bits *per source symbol*.

If X is an iid source, i.e., $X_1, X_2, \dots, X_N$ are iid according to some pmf P with entropy $H(X)$, then

$$H(X^N) = NH(X) \quad \Rightarrow \quad \frac{\mathbb{E}[\ell(X^N)]}{N} \geq H(X)$$

19 / 21

A couple of remarks:

1. Since the above holds for arbitrary N , we have shown that for an iid source with entropy $H(X)$, *any* prefix-free code has average code length that satisfies

$$\mathbb{E} \left[\frac{1}{N} \ell(X^N) \right] \geq H(X).$$

In fact the above bound holds for any uniquely decodable code since all we used in the proof was Kraft's inequality, which holds for uniquely decodable codes as well.

2. Recall that in Handout 2, we constructed a uniquely decodable binary code for blocks of length N , such that for any $\epsilon > 0$, its expected code-length satisfied

$$\mathbb{E} \left[\frac{1}{N} \ell(X^N) \right] \leq H(X) + \epsilon$$

for sufficiently large N .

20 / 21

1. and 2. together form Shannon's lossless source coding theorem for iid sources:

- ▶ You can construct a uniquely decodable code with expected length arbitrarily close to the entropy (by taking block length N large enough).
- ▶ Conversely, *you cannot* construct a uniquely decodable code with expected length less than the source entropy.

Key Q: How to construct *practical* codes with expected length close to $H(X)$?

Next we will study some practical prefix-free coding techniques.